KAVACH · GeoSatya Integration Report

Incident Intelligence + Dispatch + live tracking, ETA and ETR, integrated into GeoSatya.
Repository **https://github.com/ananyadhagat/Kavach-epam**, branch **geosatya-location-verification** (base e4e3e91), module `services/geosatya`. Generated 2026-10-30.

> **State of the work.** Implementation, tests (139 passed) and localhost verification are complete. **Nothing has been committed or pushed.** Remaining: team decisions (dispatch fallback default, ratings), a real-phone test, an optional routing provider, then commit and push.

1. Project overview

GeoSatya is KAVACH's location-verified civic reporting module. A citizen reports an issue with a photo and device location. The report is now analysed by the Incident Intelligence agent and dispatched with the Dispatch agent's ranking rules. The assigned worker can share their location while travelling, so the citizen and supervisor see an approximate arrival time (ETA) and resolution estimate (ETR). Completion is still verified on the server against the original incident location (15 m geofence), and a supervisor decides.

```
Citizen report → saved → Incident Intelligence (priority) → Dispatch (worker) → worker shares location
→ ETA + ETR → completion photo → server geofence → supervisor review → resolved (actual time recorded)
```

2. Existing GeoSatya capabilities (verified in code and tests)

- Citizen reporting: category, description, photo, automatic device location with four separate timestamps
- EXIF GPS reading and conflict flags; gallery and duplicate-photo flags; simulated GPS always labelled
- Locked incident location; audited supervisor correction keeps the original
- Server-side Haversine geofence: PASS / FAIL / REVIEW_REQUIRED / NOT_CHECKED, configurable 10–20 m
- Supervisor approve / reject / request evidence with reasons; full audit trail
- All 110 original tests still pass unchanged

3. Reused from Incident Intelligence

- Agent called over HTTP (POST /api/v1/incidents) with a stable X-Request-ID per incident
- Output stored separately in incident_intelligence; citizen category and text never overwritten
- Severity maps to dispatch priority HIGH/MEDIUM/LOW, otherwise NORMAL
- NOT_CONFIGURED, UNAVAILABLE, FAILED and CONFLICT outcomes recorded; report still dispatched; supervisor retry endpoint
- Verified end to end against the real agent running locally in mock mode (no Gemini key)

4. Reused from Dispatch Work

- dispatch_task ranking reused with three documented minimal changes
- Worker data now from GeoSatya worker_profiles and shared locations, not the branch's static in-memory list
- Configurable fallback: labelled workload assignment or escalation to supervisor

- Supervisor manual assign, reassign (reason required) and retry dispatch; one task per incident guarantees no duplicates
- The branch's own HTTP service is not called (static workers near Ujjain go stale 15 min after start)

5. ETA and live-tracking design

- Assigned worker shares location during ASSIGNED/IN_PROGRESS via browser watchPosition or labelled simulation
- Citizen sees latest location only while live; supervisor also last-known and up to 20 recent points
- Points deleted on completion submission or reassignment; updates throttled to 10 s
- Arrival estimated within 50 m; never replaces the 15 m server geofence check
- Foreground-only: a web page cannot track in the background
- Approximate only: straight-line distance x 1.4 route factor at observed approach speed or assumed 20 km/h
- Shown as a minute range with states WAITING_FOR_LOCATION, AVAILABLE, STALE, ARRIVED, NOT_APPLICABLE
- No road-routing provider configured

6. ETR and resolution-duration design

- Recomputed only on events: assignment, first location, arrival, work start, completion, rework, reassignment, resolution
- Work duration = median of at least 3 resolved jobs in the category, else category baseline in categories.json
- Actual resolution time and breakdown computed from recorded timestamps only

7. Integration decisions (confirm with the team)

- GeoSatya stays the intake and system of record; the intelligence agent runs as a separate service (merging branches would collide at the repo root)
- Dispatch's ranking rules are reused in-process; its HTTP service with static in-memory workers is not used
- DISPATCH_FALLBACK=workload by default (labelled, distance not claimed); 'escalate' follows the Dispatch agent's rule strictly
- Worker check-in shares one location for dispatch; continuous tracking only during an assigned task
- Status mapping uses existing statuses; 'accepted' and 'cancelled' are not supported; reassignment is
- Ratings not implemented: proposal in docs/INTEGRATION.md §8

8. Files and modules

Modified (18):

- services/geosatya/.env.example
- services/geosatya/README.md
- services/geosatya/frontend/js/maps.js
- services/geosatya/frontend/js/pages/dashboard.js
- services/geosatya/frontend/js/pages/incident.js
- services/geosatya/frontend/js/pages/tasks.js
- services/geosatya/frontend/styles.css

- services/geosatya/geosatya/api/routes.py
- services/geosatya/geosatya/api/schemas.py
- services/geosatya/geosatya/api/views.py
- services/geosatya/geosatya/config.py
- services/geosatya/geosatya/data/categories.json
- services/geosatya/geosatya/main.py
- services/geosatya/geosatya/models.py
- services/geosatya/geosatya/seed.py
- services/geosatya/geosatya/services/common.py
- services/geosatya/geosatya/services/incidents.py
- services/geosatya/geosatya/services/tasks.py

Added (14):

- services/geosatya/docs/INTEGRATION.md
- services/geosatya/docs/GEOSATYA_INTEGRATION_HANDOFF.md
- services/geosatya/docs/geosatya_integration_report.json
- services/geosatya/docs/GeoSatya_Integration_Report.pdf
- services/geosatya/frontend/js/live.js
- services/geosatya/geosatya/dispatch/__init__.py
- services/geosatya/geosatya/dispatch/ranking.py
- services/geosatya/geosatya/integrations/__init__.py
- services/geosatya/geosatya/integrations/intelligence.py
- services/geosatya/geosatya/services/dispatch.py
- services/geosatya/geosatya/services/estimates.py
- services/geosatya/geosatya/services/intelligence.py
- services/geosatya/geosatya/services/tracking.py
- services/geosatya/tests/test_integration.py

9. Test results (actually run)

Passed:

- 139 pytest tests passed (110 original + 29 in tests/test_integration.py), 1 dependency deprecation warning
- Smoke: intelligence COMPLETED with severity HIGH (INJURY rule) -> priority HIGH
- Smoke: dispatch ranked the 0.6 km worker over the 1.6 km worker
- Smoke: idempotent retry returned the same incident
- Smoke: ETA available, arrival detected at 20 m, geofence PASS at 4 m, tracking ended and points purged, ETR replaced by actual resolution time
- Browser: no console errors; no horizontal overflow at 375 px

Failed: none.

Not run:

- Partner branches' own test suites
- Real phone GPS and camera (the browser pane blocks them; labelled simulation used)

- Gemini mode of the intelligence agent (no API key)
- Google Maps (no key); OpenStreetMap used

10. Localhost setup

- Intelligence agent (optional): in a checkout of incident-intelligence-agent, set KAVACH_LLM_MODE=mock and run: python -m uvicorn kavach.main:app --app-dir src --port 8001
- GeoSatya: cd services/geosatya; set INTELLIGENCE_URL=http://127.0.0.1:8001 (optional); run: uvicorn geosatya.main:create_app --factory --reload
- Open http://localhost:8000 and pick a demo role; enable Simulate GPS (demo) when there is no GPS

Ports: GeoSatya 8000, Incident Intelligence agent 8001. Health: `/health` on both. Location and camera work on `http://localhost`; phones need HTTPS.

11. Known limitations and issues

- ETA has no road routing
- Browser tracking stops when the page is closed or suspended
- ETR baselines are placeholders until history exists
- Demo identities are not authentication
- incident-intelligence-agent requirements.txt is missing python-dotenv
- dispatch-work has an empty requirements.txt, app/_init_.py misnamed, and static in-memory workers
- No admin UI to set worker skills; non-demo workers start with no skills
- No accept or cancel step in the task lifecycle
- SQLite with create_all; no migrations

12. Checkpoints

#	Checkpoint	Status	What changed	Verification
1	Existing GeoSatya preserved and reviewed	Completed	Nothing removed; geofence rule untouched	110 original tests pass
2	Incident Intelligence integration	Completed	HTTP client, stored output, retry, supervisor card	9 tests + live run vs real agent (mock)
3	Dispatch integration	Completed	Reused ranking, worker profiles, dispatch records, assign/reassign/retry	8 tests + live run
4	ETA and live tracking	Completed	Pings, arrival, privacy, worker sharing UI	8 tests + browser check
5	ETR and resolution monitoring	Completed	Event-based ETR, historical median, actual durations	4 tests + live run
6	Frontend integration	Completed	live.js; incident, task, dashboard pages	Browser desktop + 375 px
7	Backend and database integration	Completed	6 additive tables, config, .env.example	Full suite
8	Tests and bug fixes	Completed	tests/test_integration.py	139 passed
9	Localhost verification	Completed	GeoSatya :8000 + agent :8001	Health OK; end-to-end flow passed
10	Documentation and remaining work	Partly done	README, INTEGRATION.md, handoff, JSON, this PDF	Ratings, routing, real phone, commit/push pending

13. Exact continuation instructions

1. Open `D:\Desktop\Kavach-epam-team`, run `git status`, and confirm the branch is `geosatya-location-verification` and the changes are only under `services/geosatya/`.
2. `cd services\geosatya` then `python -m pytest`; expect 139 passed.
3. Run the app (and optionally the agent) as in section 10; follow the "Integrated demo" in the README.
4. Decide the open questions in section 7, then commit using explicit paths (not `git add .`) and push to `origin geosatya-location-verification`.
5. Next features: ratings (proposal ready), a routing provider for road-distance ETA, an admin screen for worker skills, and a real-phone HTTPS test.

14. Final JSON report


```
{
  "project": "Kavach GeoSatya Integration",
  "generated": "2026-10-30",
  "repository": {
    "url": "https://github.com/ananyadhagat/Kavach-epam",
    "branch": "geosatya-location-verification",
    "base_commit": "e4e3e91",
    "module_path": "services/geosatya"
  },
  "sources_inspected": {
    "geosatya": "origin/geosatya-location-verification e4e3e91",
    "incident_intelligence_agent": "origin/incident-intelligence-agent e2ff0e9 (ZIP Kavach-epam-incident-intelligence-agent.zip
has identical code)",
    "dispatch_work": "origin/dispatch-work 6ef3bf9 (ZIP Kavach-epam-dispatch-work.zip has identical code)"
  },
  "integration": {
    "geosatya": {
      "status": "completed",
      "details": [
        "Existing reporting, EXIF, geofence, review and audit preserved; all 110 original tests still pass",
        "Report is committed before any other agent is called",
        "Optional Idempotency-Key on POST /incidents prevents duplicate reports on retry",
        "Additive schema only: 6 new tables, no existing table or column changed"
      ]
    },
    "incident_intelligence": {
      "status": "completed",
      "details": [
        "Agent called over HTTP (POST /api/v1/incidents) with a stable X-Request-ID per incident",
        "Output stored separately in incident_intelligence; citizen category and text never overwritten",
        "Severity maps to dispatch priority HIGH/MEDIUM/LOW, otherwise NORMAL",
        "NOT_CONFIGURED, UNAVAILABLE, FAILED and CONFLICT outcomes recorded; report still dispatched; supervisor retry endpoint",
        "Verified end to end against the real agent running locally in mock mode (no Gemini key)"
      ]
    },
    "dispatch_work": {
      "status": "completed",
      "details": [
        "dispatch_task ranking reused with three documented minimal changes",
        "Worker data now from GeoSatya worker_profiles and shared locations, not the branch's static in-memory list",
        "Configurable fallback: labelled workload assignment or escalation to supervisor",
        "Supervisor manual assign, reassign (reason required) and retry dispatch; one task per incident guarantees no duplicates",
        "The branch's own HTTP service is not called (static workers near Ujjain go stale 15 min after start)"
      ]
    },
    "eta": {
      "status": "completed",
      "details": [
        "Approximate only: straight-line distance x 1.4 route factor at observed approach speed or assumed 20 km/h",
```

```
        "Shown as a minute range with states WAITING_FOR_LOCATION, AVAILABLE, STALE, ARRIVED, NOT_APPLICABLE",
        "No road-routing provider configured"
      ]
    },
    "live_worker_tracking": {
      "status": "completed",
      "details": [
        "Assigned worker shares location during ASSIGNED/IN_PROGRESS via browser watchPosition or labelled simulation",
        "Citizen sees latest location only while live; supervisor also last-known and up to 20 recent points",
        "Points deleted on completion submission or reassignment; updates throttled to 10 s",
        "Arrival estimated within 50 m; never replaces the 15 m server geofence check",
        "Foreground-only: a web page cannot track in the background"
      ]
    },
    "etr_resolution_estimation": {
      "status": "completed",
      "details": [
        "Recomputed only on events: assignment, first location, arrival, work start, completion, rework, reassignment,
resolution",
        "Work duration = median of at least 3 resolved jobs in the category, else category baseline in categories.json",
        "Actual resolution time and breakdown computed from recorded timestamps only"
      ]
    },
    "ratings": {
      "status": "not_started",
      "details": [
        "No rating exists in any inspected branch; minimal proposal in docs/INTEGRATION.md section 8 awaiting a team decision"
      ]
    }
  },
  "testing": {
    "tests_run": [
      "pytest services/geosatya (full suite)",
      "HTTP smoke test against running GeoSatya :8000 and Incident Intelligence agent :8001 (mock mode)",
      "Browser check of citizen, worker and supervisor pages at desktop and 375 px width"
    ],
    "passed": [
      "139 pytest tests passed (110 original + 29 in tests/test_integration.py), 1 dependency deprecation warning",
      "Smoke: intelligence COMPLETED with severity HIGH (INJURY rule) -> priority HIGH",
      "Smoke: dispatch ranked the 0.6 km worker over the 1.6 km worker",
      "Smoke: idempotent retry returned the same incident",
      "Smoke: ETA available, arrival detected at 20 m, geofence PASS at 4 m, tracking ended and points purged, ETR replaced by
actual resolution time",
      "Browser: no console errors; no horizontal overflow at 375 px"
    ],
    "failed": [],
    "not_run": [
      "Partner branches' own test suites",
      "Real phone GPS and camera (the browser pane blocks them; labelled simulation used)",
      "Gemini mode of the intelligence agent (no API key)",
      "Google Maps (no key); OpenStreetMap used"
    ],
    "limitations": [
      "ETA has no road routing",
      "Browser tracking stops when the page is closed or suspended",
      "ETR baselines are placeholders until history exists",
      "Demo identities are not authentication"
    ]
  },
  "localhost": {
    "status": "verified",
    "startup_instructions": [
      "Intelligence agent (optional): in a checkout of incident-intelligence-agent, set KAVACH_LLM_MODE=mock and run: python -m
uvicorn kavach.main:app --app-dir src --port 8001",
      "GeoSatya: cd services/geosatya; set INTELLIGENCE_URL=http://127.0.0.1:8001 (optional); run: uvicorn
geosatya.main:create_app --factory --reload",
      "Open http://localhost:8000 and pick a demo role; enable Simulate GPS (demo) when there is no GPS"
    ],
```

```
    "required_services": [
      "GeoSatya FastAPI app (also serves the frontend)",
      "Incident Intelligence agent (optional)"
    ],
    "ports": [
      8000,
      8001
    ]
  },
  "git": {
    "files_changed": [
      "services/geosatya/.env.example",
      "services/geosatya/README.md",
      "services/geosatya/frontend/js/maps.js",
      "services/geosatya/frontend/js/pages/dashboard.js",
      "services/geosatya/frontend/js/pages/incident.js",
      "services/geosatya/frontend/js/pages/tasks.js",
      "services/geosatya/frontend/styles.css",
      "services/geosatya/geosatya/api/routes.py",
      "services/geosatya/geosatya/api/schemas.py",
      "services/geosatya/geosatya/api/views.py",
      "services/geosatya/geosatya/config.py",
      "services/geosatya/geosatya/data/categories.json",
      "services/geosatya/geosatya/main.py",
      "services/geosatya/geosatya/models.py",
      "services/geosatya/geosatya/seed.py",
      "services/geosatya/geosatya/services/common.py",
      "services/geosatya/geosatya/services/incidents.py",
      "services/geosatya/geosatya/services/tasks.py"
    ],
    "files_added": [
      "services/geosatya/docs/INTEGRATION.md",
      "services/geosatya/docs/GEOSATYA_INTEGRATION_HANDOFF.md",
      "services/geosatya/docs/geosatya_integration_report.json",
      "services/geosatya/docs/GeoSatya_Integration_Report.pdf",
      "services/geosatya/frontend/js/live.js",
      "services/geosatya/geosatya/dispatch/__init__.py",
      "services/geosatya/geosatya/dispatch/ranking.py",
      "services/geosatya/geosatya/integrations/__init__.py",
      "services/geosatya/geosatya/integrations/intelligence.py",
      "services/geosatya/geosatya/services/dispatch.py",
      "services/geosatya/geosatya/services/estimates.py",
      "services/geosatya/geosatya/services/intelligence.py",
      "services/geosatya/geosatya/services/tracking.py",
      "services/geosatya/tests/test_integration.py"
    ],
    "commit_created": false,
    "push_completed": false,
    "target_branch": "geosatya-location-verification"
  },
  "checkpoints": [
    {
      "id": 1,
      "name": "Existing GeoSatya preserved and reviewed",
      "status": "completed"
    },
    {
      "id": 2,
      "name": "Incident Intelligence integration",
      "status": "completed"
    },
    {
      "id": 3,
      "name": "Dispatch integration",
      "status": "completed"
    },
    {
      "id": 4,
```

```
      "name": "ETA and live tracking",
      "status": "completed"
    },
    {
      "id": 5,
      "name": "ETR and resolution monitoring",
      "status": "completed"
    },
    {
      "id": 6,
      "name": "Frontend integration",
      "status": "completed"
    },
    {
      "id": 7,
      "name": "Backend and database integration",
      "status": "completed"
    },
    {
      "id": 8,
      "name": "Tests and bug fixes",
      "status": "completed"
    },
    {
      "id": 9,
      "name": "Localhost verification",
      "status": "completed"
    },
    {
      "id": 10,
      "name": "Documentation and remaining work",
      "status": "partially_completed",
      "note": "Docs done; ratings, routing, real-device test and commit/push pending"
    }
  ],
  "known_issues": [
    "incident-intelligence-agent requirements.txt is missing python-dotenv",
    "dispatch-work has an empty requirements.txt, app/_init_.py misnamed, and static in-memory workers",
    "No admin UI to set worker skills; non-demo workers start with no skills",
    "No accept or cancel step in the task lifecycle",
    "SQLite with create_all; no migrations"
  ],
  "next_steps": [
    "Review the diff, then commit to geosatya-location-verification and push to origin (not done; awaiting approval)",
    "Team decision on DISPATCH_FALLBACK default and on the rating proposal",
    "Test on a real phone over HTTPS",
    "Optionally add a routing provider for road-distance ETA",
    "Agree with the intelligence owner whether its classification should replace the citizen's category"
  ],
  "handoff_document": "services/geosatya/docs/GEOSATYA_INTEGRATION_HANDOFF.md"
}
```